

Hands-on Activity 4.5 Web Accessibility with CSS Lab Project	
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1. Discussion	
For this module 9 it focuses on the principles of web accessibility. This teaches the different ways or elements in which we can make the website we create usable for everyone which includes people with disabilities. This helps in emphasizing the ethical and practical aspects of web design in which we should include our scope to users who rely on assistive technologies. The key aspect of accessibility is the implementation of a clear and consistent styling which lessens confusion and enables predictable navigation for users. Interactive elements like buttons and links should be easily distinguishable from static content to reduce cognitive load. Also visual aspects such as color contrast and text attributes should be able to follow WCAG guidelines. Lastly is the CSS for screen readers, teaching how to make hidden content accessible and ensuring that the visual presentation aligns with the logical reading order.	
3. Procedure	
9.6.1 Web Accessibility with CSS Lab Project Scenario You work for a company that is launching a new website. Your task is to ensure the website is accessible to all users, including those with disabilities. Use CSS to enhance the site's accessibility by implementing clear navigation, identifiable links, simple and predictable layouts, readable text, and screen reader accessibility. Instructions Follow the steps below to create and style the web page. 1. Clear Navigation Create a navigation menu that is easy to use and accessible. Ensure the menu items are clearly separated and provide visual feedback when hovered over. Create a navigation bar with list items and links. Style the navigation menu to remove default list styling, add spacing between items, and provide visual feedback on hover. Show Sample Implementation 2. Identifiable Links Ensure all links are easily identifiable by using distinct colors and underline styles. This helps users, especially those with visual impairments, to distinguish links from regular text. Set a distinct color and underline for links. Provide visual feedback by changing the color on hover and focus. Show Sample Implementation 3. Simple and Predictable Layouts Design the page layout to be simple and predictable. This helps users with cognitive disabilities	

understand and navigate the site more easily.

Create a basic layout with a sidebar and main content area.

Use flexbox to structure the layout.

Show Sample Implementation

4. Readable Text

Ensure your text is readable by setting appropriate font sizes and maintaining high contrast between text and background colors.

Set the font size of 16px and color of #333 for body text.

Ensure high contrast between body text and background colors by setting the background-color property to #f4f4f4.

Show Sample Implementation

5. Screen Reader Accessibility

Ensure that screen reader users can access all important content by hiding it visually while keeping it accessible to screen readers.

Create a class named screen-reader-text to hide content visually but make it accessible to screen readers. In your index.html, use this class to provide additional context or instructions for screen reader users. For example, add a visually hidden label for a menu button.

Use the aria-labelledby attribute to associate the button with the hidden label.

4. Output

HTML CODE

HTML

```
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width,
initial-scale=1.0">
  <title>Web Accessibility with CSS</title>
  <link rel="stylesheet" href="styles.css">
</head>
<body>
  <header class="header">
    <h1>Accessible Web Design</h1>
  </header>
  <nav>
    <ul class="navigation-menu">
      <li><a href="/">Home</a></li>
      <li><a href="/about">About</a></li>
      <li><a href="/services">Services</a></li>
      <li><a href="/contact">Contact</a></li>
    </ul>
  </nav>
  <div class="content">
    <aside class="sidebar">
```

```

        Sidebar content
    </aside>
    <main class="main-content">
        <section class="about">
            <h2>About Us</h2>
            <p>We are dedicated to creating accessible web
designs.</p>
        </section>
        <section class="features">
            <h2>Features</h2>
            <ul class="feature-list">
                <li>Clear Navigation</li>
                <li>Identifiable Links</li>
                <li>Simple Layouts</li>
                <li>Readable Text</li>
                <li>Screen Reader Accessibility</li>
            </ul>
            <p>For more information, visit our <a
href="https://example.com">Help Center</a>.</p>
        </section>
        <section class="example">
            <h2>Example Text</h2>
            <p>This is standard text with a font size of 16px. It
should be clear and easy to read.</p>
            <p class="large-text">This text is larger for better
visibility.</p>
            <div class="high-contrast-text">
                This is an example of high contrast text. Black text
on a white background.
            </div>
        </section>
    </main>
</div>
<footer class="footer">
    <p>&copy; 2050 Accessible Web Design</p>
</footer>
</body>
</html>

```

CSS

```

CSS
/* styles.css */

body {
    color: #333;
    background-color: #f4f4f4;
}

```

```
    font-size: 16px;
  }

  .navigation-menu {
    list-style: none;
    padding: 0;
  }
  .navigation-menu li {
    display: inline;
    margin-right: 20px;
  }
  .navigation-menu a {
    text-decoration: none;
    color: #0654b3;
  }
  .navigation-menu a:hover {
    text-decoration: underline;
  }

  a {
    color: #0654b3;
    text-decoration: underline;
  }
  a:hover,
  a:focus {
    color: #002766;
  }

  .content {
    display: flex;
    flex-direction: row;
  }
  .sidebar {
    flex: 1;
    order: 1;
  }
  .main-content {
    flex: 2;
    order: 2;
  }

  .large-text {
    font-size: 20px;
  }
  .high-contrast-text {
    color: #000;
    background-color: #fff;
  }

  .screen-reader-text {
```

```
position: absolute;
width: 1px;
height: 1px;
margin: -1px;
padding: 0;
overflow: hidden;
clip: rect(0, 0, 0, 0);
border: 0;
}
```

OUTPUT

Accessible Web Design

[Home](#) [About](#) [Services](#) [Contact](#)

Sidebar content

About Us

We are dedicated to creating accessible web designs.

Features

- Clear Navigation
- Identifiable Links
- Simple Layouts
- Readable Text
- Screen Reader Accessibility

For more information, visit our [Help Center](#).

Example Text

This is standard text with a font size of 16px. It should be clear and easy to read.

This text is larger for better visibility.

This is an example of high contrast text. Black text on a white background.

© 2020 Accessible Web Design

5. Supplementary Activity

HTML CODE

```
HTML
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width,
initial-scale=1.0">
  <title>Accessible Web Design</title>
  <link rel="stylesheet" href="supple.css">
</head>

<body>
```

```

<a href="#main-content" class="skip-link">Skip to Main Content</a>

<header class="header" role="banner">
  <h1>Accessible Web Design</h1>
</header>

<nav role="navigation" aria-label="Main Navigation">
  <ul class="navigation-menu">
    <li><a href="/" aria-current="page">Home</a></li>
    <li><a href="/about">About</a></li>
    <li><a href="/services">Services</a></li>
    <li><a href="/contact">Contact</a></li>
  </ul>
</nav>

  <div class="content">
    <aside class="sidebar">
      <nav aria-labelledby="sidebar-heading">
        <h2 id="sidebar-heading">Accessibility Topics</h2>
        <ul class="resource-list">
          <li><a href="#semantic-html">Semantic
HTML</a></li>
          <li><a href="#aria">ARIA Roles</a></li>
          <li><a href="#color-contrast">Color
Contrast</a></li>
          <li><a href="#keyboard">Keyboard
Navigation</a></li>
          <li><a href="#screen-readers">Screen
Readers</a></li>
          <li><a href="#responsive">Responsive
Design</a></li>
        </ul>
      </nav>
    </aside>

    <main id="main-content" class="main-content" role="main">

      <section class="about">
        <h2>About Us</h2>

        <figure>
          
          <figcaption>
            Our team working together to build inclusive
digital experiences.
          </figcaption>
        </figure>

```

```

        <p>
            We are dedicated to creating inclusive and
            accessible web designs that work for everyone.
        </p>
    </section>

    <section class="features">
        <h2>Accessibility Features</h2>
        <ul class="feature-list">
            <li>Clear Navigation</li>
            <li>Identifiable Links</li>
            <li>Simple Layouts</li>
            <li>Readable Text</li>
            <li>Screen Reader Compatibility</li>
        </ul>
        <p>
            For more information, visit our
            <a href="https://example.com" target="_blank"
rel="noopener">
                Help Center (opens in new tab)
            </a>.
        </p>
    </section>

    <section class="example">
        <h2>Readable Text Example</h2>
        <p>This is standard text with a readable font size and
line height.</p>
        <p class="large-text">This text is larger for improved
visibility.</p>
        <div class="high-contrast-text">
            This is high contrast text: black on white for
            better readability.
        </div>
    </section>

    </main>
</div>

<footer class="footer" role="contentinfo">
    <p>&copy; 2050 Accessible Web Design</p>
</footer>

</body>
</html>

```

CSS CODE

CSS

```
body {  
  font-family: Arial, Helvetica, sans-serif;  
  font-size: 16px;  
  line-height: 1.6;  
  margin: 0;  
  background-color: #f5f7fa;  
  color: #222;  
}
```

```
:focus {  
  outline: 3px solid #ffcc00;  
  outline-offset: 3px;  
}
```

```
img {  
  max-width: 100%;  
  height: auto;  
  display: block;  
}
```

```
.skip-link {  
  position: absolute;  
  top: -40px;  
  left: 0;  
  background: #000;  
  color: #fff;  
  padding: 8px 16px;  
  z-index: 100;  
  text-decoration: none;  
}
```

```
.skip-link:focus {  
  top: 0;  
}
```

```
.header {  
  background-color: #003366;  
  color: white;  
  padding: 1.5rem;  
  text-align: center;  
}
```

```
nav {  
  background-color: #0055a5;
```



```
}

.navigation-menu {
  display: flex;
  list-style: none;
  margin: 0;
  padding: 0;
  justify-content: center;
}

.navigation-menu li {
  margin: 0;
}

.navigation-menu a {
  display: block;
  padding: 1rem 1.5rem;
  color: white;
  text-decoration: none;
  font-weight: bold;
}

.navigation-menu a:hover,
.navigation-menu a:focus {
  background-color: #002244;
  outline: 3px solid #ffcc00;
}

.content {
  display: flex;
  flex-wrap: nowrap;
  gap: 2rem;
  padding: 2rem;
}

.sidebar {
  flex: 0 0 250px;
  background-color: #ffffff; /* white background */
  padding: 1.5rem;
  border-radius: 12px;
  box-shadow: 0 4px 12px rgba(0,0,0,0.08);
  height: fit-content;
}

.sidebar h2 {
  background: none;
  color: #ffffff;
}
```

```

        font-size: 1.25rem;
        border-bottom: 2px solid #0055a5;
        padding-bottom: 0.3rem;
        margin-bottom: 1rem;
    }

    .resource-list {
        list-style: none;
        padding: 0;
        margin: 0;
    }

    .resource-list li {
        margin-bottom: 0.5rem;
    }

    /* Style the links */
    .resource-list a {
        display: block;
        padding: 0.5rem 1rem;
        background-color: #0055a5;
        color: #ffffff;
        text-decoration: none;
        border-radius: 6px;
        transition: background-color 0.2s ease;
    }

    .resource-list a:hover {
        background-color: #003366;
    }

    .main-content {
        flex: 1;
    }

    @media (max-width: 768px) {
        .content {
            flex-direction: column;
        }

        .sidebar {
            flex: 1 1 100%;
            margin-bottom: 1.5rem;
        }
    }

    section {
        margin-bottom: 2rem;
    }

```

```
    background: #ffffff;
    padding: 1.5rem;
    border-radius: 10px;
    box-shadow: 0 3px 8px rgba(0,0,0,0.05);
}

.feature-list {
    padding-left: 20px;
}

.large-text {
    font-size: 1.25rem;
    font-weight: bold;
}

.high-contrast-text {
    background-color: #ffffff;
    color: #000000;
    padding: 1rem;
    border: 2px solid #000;
    margin-top: 1rem;
}

a {
    color: #0055a5;
}

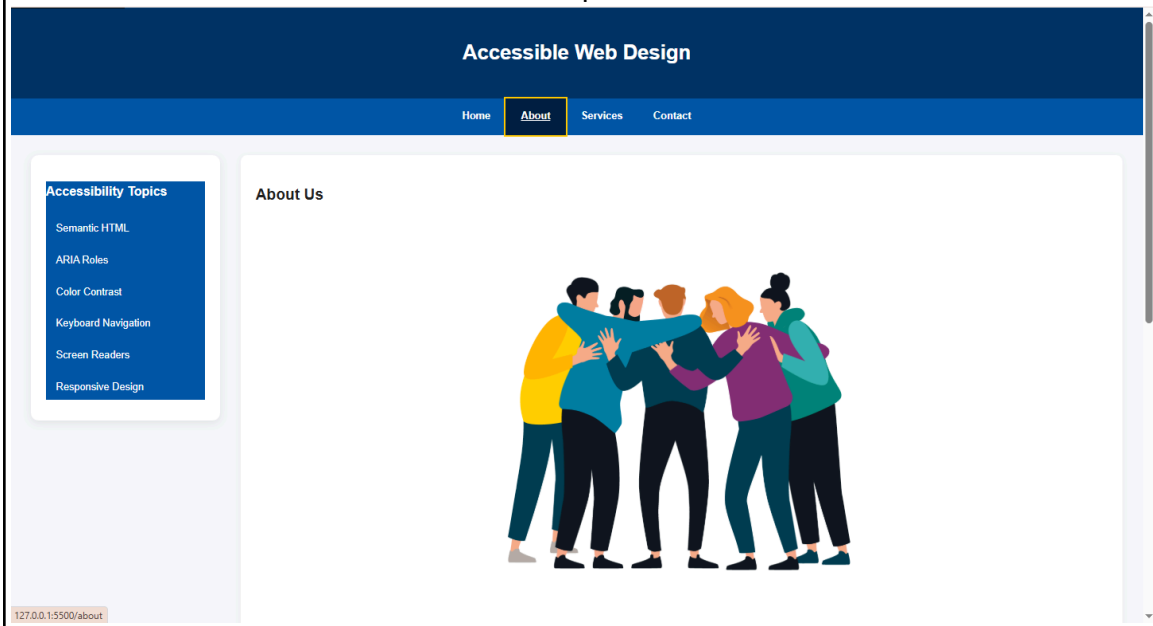
a:hover,
a:focus {
    text-decoration: underline;
    outline: 2px solid #ffcc00;
}

.footer {
    background-color: #003366;
    color: white;
    text-align: center;
    padding: 1rem;
}

@media (max-width: 768px) {
    .content {
        flex-direction: column;
    }
}
```

```
.navigation-menu {  
  flex-direction: column;  
  align-items: center;  
}  
}
```

Output



Accessibility Topics

Semantic HTML

ARIA Roles

Color Contrast

Keyboard Navigation

Screen Readers

Responsive Design

Our team working together to build inclusive digital experiences.

We are dedicated to creating inclusive and accessible web designs that work for everyone.

Accessibility Features

- Clear Navigation
- Identifiable Links
- Simple Layouts
- Readable Text
- Screen Reader Compatibility

For more information, visit our [Help Center \(opens in new tab\)](#).

Readable Text Example

This is standard text with a readable font size and line height.

This text is larger for improved visibility.

This is high contrast text: black on white for better readability.

After performing this activity I was able to learn about Web Accessibility. In this activity I learned that we should aim to build websites or applications that can also cater the needs of users that need assistive technologies. The different layout practices and WCAG guidelines can help in promoting accessibility for such users. With that in mind, I tried to implement these practices on the sample website. I did not add much content to the initial website but only tried to use as much of the different teachings of accessibility on the website. On the website I focused primarily on fixing the visual aspects by using clear contrasting colors and fixing the text presentations. Also I used a simple flow of the website and implemented alternative text for the image which can help other users with disabilities. Overall, this activity helped me learn principles of accessibility in web development which can be useful for improving my projects to also cater users with disabilities.